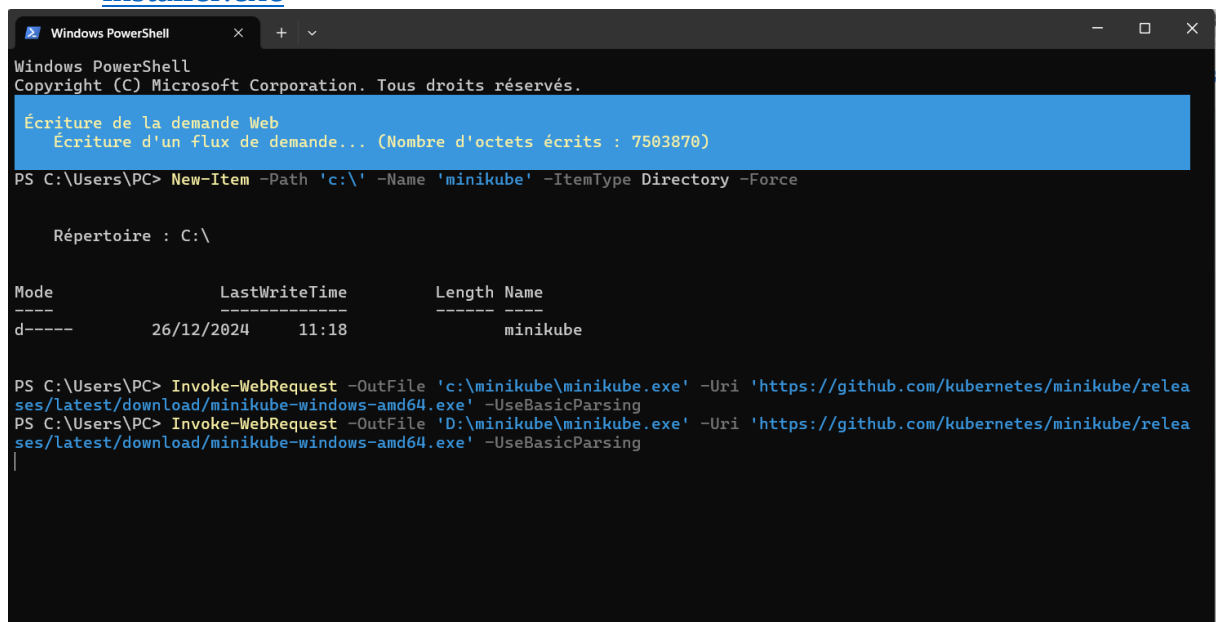


Atelier 11: mise en œuvre de Kubernetes
Installation et configuration de minikube
Master DSBD & IA

Pour cet atelier nous allons travailler avec **Minikube** et **Kubectl**. Minikube est un **outil facilitant l'exécution locale de Kubernetes**. Minikube exécute un cluster Kubernetes à noeud unique dans une machine virtuelle (VM) de votre ordinateur portable et on a besoin du conteneur Docker (ou similaire) ou un environnement de machine virtuelle comme virtualbox, vmware workstation. Kubectl est un **outil en ligne de commande pour contrôler des clusters Kubernetes**.

1. Installation de minikube

- ✓ Télécharger :
- ✓ <https://github.com/kubernetes/minikube/releases/latest/download/minikube-windows-amd64.exe>
- ✓ <https://storage.googleapis.com/minikube/releases/latest/minikube-installer.exe>



```
Windows PowerShell
Copyright (C) Microsoft Corporation. Tous droits réservés.

Écriture de la demande Web
Écriture d'un flux de demande... (Nombre d'octets écrits : 7503870)

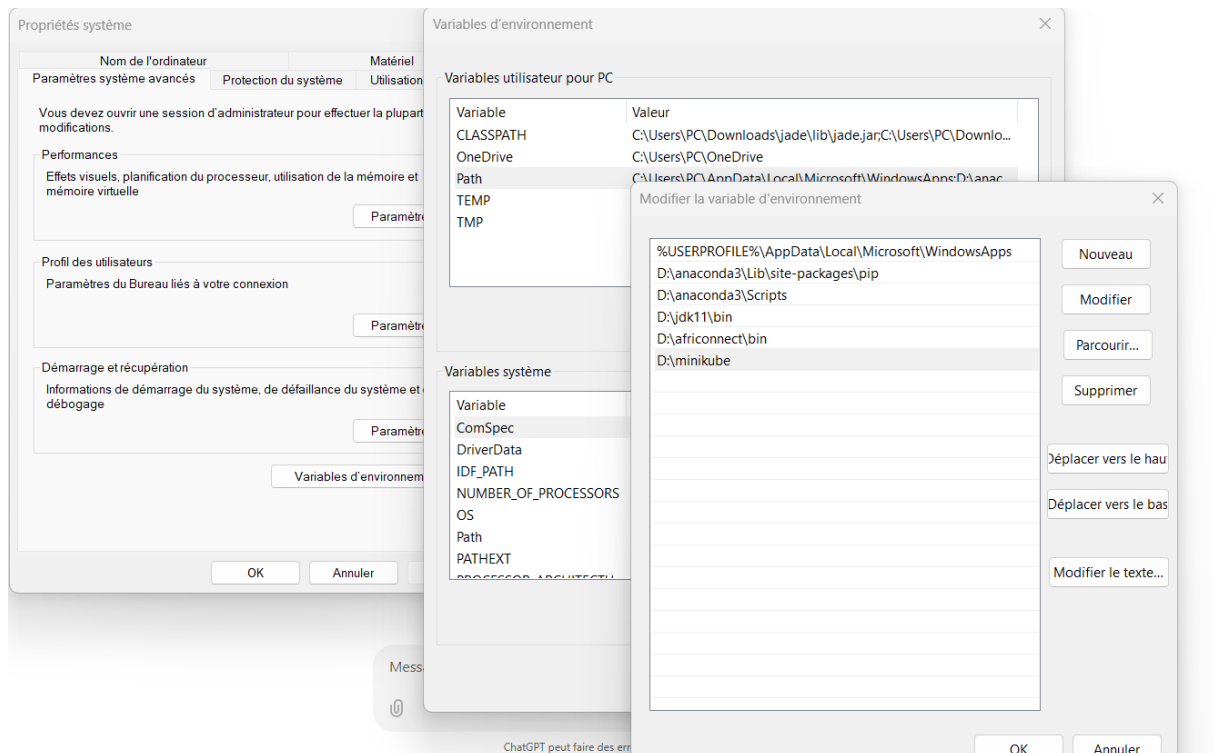
PS C:\Users\PC> New-Item -Path 'c:\' -Name 'minikube' -ItemType Directory -Force

Répertoire : C:\

Mode                LastWriteTime         Length Name
----                -
d-----         26/12/2024   11:18             minikube

PS C:\Users\PC> Invoke-WebRequest -OutFile 'c:\minikube\minikube.exe' -Uri 'https://github.com/kubernetes/minikube/releases/latest/download/minikube-windows-amd64.exe' -UseBasicParsing
PS C:\Users\PC> Invoke-WebRequest -OutFile 'D:\minikube\minikube.exe' -Uri 'https://github.com/kubernetes/minikube/releases/latest/download/minikube-windows-amd64.exe' -UseBasicParsing
```

- ✓ Ajouter minikube.exe binary à votre PATH



2. Créer un cluster

Avant de démarrer un cluster avec un seul node, fermer le PowerShell et réouvrez le.

minikube start

```

Administrateur : Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. Tous droits réservés.

Installez la dernière version de PowerShell pour de nouvelles fonctionnalités et améliorations ! https://aka.ms/PSWindows
PS C:\WINDOWS\system32> minikube start
* minikube v1.34.0 sur Microsoft Windows 11 Pro 10.0.26100.2605 Build 26100.2605
* Choix automatique du pilote hyperv
* Téléchargement de l'image de démarrage de la VM...
  > minikube-v1.34.0-amd64.iso....: 65 B / 65 B [-----] 100.00% ? p/s 0s
  > minikube-v1.34.0-amd64.iso: 333.55 MiB / 333.55 MiB 100.00% 17.10 MiB p
* Démarrage du nœud "minikube" primary control-plane dans le cluster "minikube"
* Téléchargement du préchargement de Kubernetes v1.31.0...
  > preloaded-images-k8s-v18-v1...: 326.69 MiB / 326.69 MiB 100.00% 19.79 M
* Création de VM hyperv (CPUs=2, Mémoire=4000MB, Disque=20000MB)...
! StartHost a échoué, mais va réessayer : creating host: create: Hyper-V PowerShell Module is not available
* Création de VM hyperv (CPUs=2, Mémoire=4000MB, Disque=20000MB)...
* Échec du démarrage de hyperv VM. L'exécution de "minikube delete" peut résoudre le problème : creating host: create: p
recreate: Hyper-V PowerShell Module is not available

X Fermeture en raison de PR_HYPERV_MODULE_NOT_INSTALLED : Failed to start host: creating host: create: precreate: Hyper-
V PowerShell Module is not available
* Suggestion : Exécutez : 'Enable-WindowsOptionalFeature -Online -FeatureName Microsoft-Hyper-V-Tools-All -All'
* Documentation: https://www.altaro.com/hyper-v/install-hyper-v-powershell-module/
* Problème connexe: https://github.com/kubernetes/minikube/issues/9040
PS C:\WINDOWS\system32>
  
```

Si jamais un problème démarrage lié à Hyper-V, utiliser la commande suivante :

Enable-WindowsOptionalFeature -Online -FeatureName Microsoft-Hyper-V-Tools-All -All

3. Interagir avec le cluster

Télécharger **kubectl** : <https://dl.k8s.io/release/v1.26.0/bin/windows/amd64/kubectl.exe>

Informations sur le cluster

Commandes pour manipuler le cluster :

kubectl cluster.info

```
Administrateur : Windows PowerShell
PS C:\WINDOWS\system32> kubectl cluster-info
Kubernetes control plane is running at https://172.17.225.86:8443
CoreDNS is running at https://172.17.225.86:8443/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
PS C:\WINDOWS\system32>
```

Lister les nodes (MV), ici on a 1 seul node

```
Administrateur : Windows PowerShell
PS C:\WINDOWS\system32> kubectl get nodes
NAME        STATUS    ROLES         AGE   VERSION
minikube    Ready    control-plane  18m   v1.31.0
PS C:\WINDOWS\system32>
```

4. Création d'un local registry

Dans cet atelier, nous allons créer un Registry local et y mettre nos images au lieu d'utiliser docker hub.

1. Vérifier que Docker est bien installé

```
Sélection Administrateur : Invite de commandes
C:\Windows\System32>docker -v
Docker version 27.3.1, build ce12230
```

2. Vérifier que Kube est bien installé :

```
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> kubectl version
Client Version: v1.30.5
Kustomize Version: v5.0.4-0.20230601165947-6ce0bf390ce3
Server Version: v1.31.0
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app>
```

3. Exécuter les commandes :

docker run -d -p 5000:5000 --name local-registry registry:2
minikube addons enable registry

```
PS C:\windows\system32> docker run -d -p 5000:5000 --name local-registry registry:2
Unable to find image 'registry:2' locally
2: Pulling from library/registry
813676e291ef: Download complete
dc2fb7dcec61: Download complete
dc0decf4841d: Download complete
6cb0aa443e23: Download complete
916205650bfe: Download complete
Digest: sha256:543dade69668e02e5768d7ea2b0aa4fae6aa7384c9a5a8dbecc2be5136079ddb
Status: Downloaded newer image for registry:2
1ce0781fb24b57ddb0175f79d94b5a2853b3c6a645e55c108cf7b47ed7858e78
```

4. Vérifier que le registre a bien été créé.

```
PS C:\windows\system32> docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
1ce0781fb24b	registry:2	"/entrypoint.sh /etc..."	18 seconds ago	Up 17 seconds	0.0.0.0:5000->5000/tcp	local-registry

6. Dockeriser l'application

a) Allez dans le répertoire où se trouve votre application et dockerfile.

```
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> dir
```

Répertoire : C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app

Mode	LastWriteTime	Length	Name
d----	10/12/2024 13:40		backend
d----	10/12/2024 13:40		fonts
-a----	27/10/2020 16:53	22	.gitignore
-a----	27/10/2020 16:53	1239	app.js
-a----	27/10/2020 16:53	127	Dockerfile
-a----	27/10/2020 16:53	1826	index.html
-a----	27/10/2020 16:53	1131	LICENSE
-a----	27/10/2020 16:53	572	package.json
-a----	27/10/2020 16:53	888	readme.md
-a----	27/10/2020 16:53	1071	server.js
-a----	27/10/2020 16:53	1227	site.css

b) Construire le package

Docker build -t firstapp -f Dockerfile .
docker images

```
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> Docker build -t firstapp -f Dockerfile .
ERROR: "docker buildx build" requires exactly 1 argument.
See 'docker buildx build --help'.
```

Usage: docker buildx build [OPTIONS] PATH | URL | -

Start a build

```
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> Docker build -t firstapp -f Dockerfile .
[+] Building 19.2s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> transferring dockerfile: 164B
=> [internal] load metadata for docker.io/library/node:current-slim
=> [internal] load .dockerignore
=> transferring context: 2B
=> [1/5] FROM docker.io/library/node:current-slim@sha256:7fdf54a2a5dc734af5b447021fdcc8c1c38a82c021a5c338efa0a3
=> resolve docker.io/library/node:current-slim@sha256:7fdf54a2a5dc734af5b447021fdcc8c1c38a82c021a5c338efa0a3
=> sha256:b7c0a2c9ac8d2826a32f0d8c5475612680d117d221ee9d1953c21d7dc61aa37d 448B / 448B
=> sha256:775e6820e2ab2aa4594adf6d6d165b7eb5927328d8a4f759712009166c1b2143 1.71MB / 1.71MB
=> sha256:1929817bc7d8a75b8c36401b0b85916e814ad729a5b36136fb4d65b423a25a6a 49.70MB / 49.70MB
=> sha256:c919ce427c9480520bd4b0fa00f47e9dbb0470c199edc2658139310addb2517 3.31kB / 3.31kB
=> sha256:fd674058f8f8cfa7fb8a20c006fc0128541cbbad7f7f728df570d08f9e4d92 28.23MB / 28.23MB
=> extracting sha256:fd674058f8f8cfa7fb8a20c006fc0128541cbbad7f7f728df570d08f9e4d92
=> extracting sha256:c919ce427c9480520bd4b0fa00f47e9dbb0470c199edc2658139310addb2517
=> extracting sha256:1929817bc7d8a75b8c36401b0b85916e814ad729a5b36136fb4d65b423a25a6a
=> extracting sha256:775e6820e2ab2aa4594adf6d6d165b7eb5927328d8a4f759712009166c1b2143
=> extracting sha256:b7c0a2c9ac8d2826a32f0d8c5475612680d117d221ee9d1953c21d7dc61aa37d
=> [internal] load build context
=> transferring context: 518B
=> [2/5] WORKDIR /usr/src/app
=> [3/5] COPY package.json .
=> [4/5] RUN npm install
=> [5/5] COPY . .
=> exporting to image
=> exporting layers
=> exporting manifest sha256:4e21db0b3aa041d514c5b52a0fb453222433b509f6d58ba844c631c2620719a
=> exporting config sha256:6ad1bf5b171d3add4afb40803df615d22fbc9d3339ccbfec37e87853a71410
=> exporting attestation manifest sha256:00253fe5d1768ba8ee969f47a1e62b069c34535491a8bb6dc340cb850ec6f8be
=> exporting manifest list sha256:3ff46c40d2e19098eb35b9589b0c91576b0968d70576ebdacffca365881ce4a1
=> naming to docker.io/library/firstapp:latest
=> unpacking to docker.io/library/firstapp:latest
```

View build details: [docker-desktop://dashboard/build/desktop-linux/desktop-linux/c5p0p15xvprv5xxadibjokvx](https://dashboard/build/desktop-linux/desktop-linux/c5p0p15xvprv5xxadibjokvx)

Vérifier :

```
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
firstapp	latest	3ff46c40d2e1	33 seconds ago	373MB
faouziabenabbou/firstapp	latest	a9f050b924e6	2 weeks ago	369MB
faouziabenabbou/firstappv0.1	latest	a9f050b924e6	2 weeks ago	369MB
registry	2	543dade69668	15 months ago	37.2MB
hello-world	latest	305243c73457	20 months ago	24.4kB

c) Tagger l'application dans le registre local

```
docker tag firstapp localhost:5000/firstapp:local
```

```
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> docker tag firstapp localhost:5000/firstapp:local
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
localhost:5000/firstapp	local	3ff46c40d2e1	4 minutes ago	373MB
firstapp	latest	3ff46c40d2e1	4 minutes ago	373MB
faouziabenabbou/firstapp	latest	a9f050b924e6	2 weeks ago	369MB
faouziabenabbou/firstappv0.1	latest	a9f050b924e6	2 weeks ago	369MB
registry	2	543dade69668	15 months ago	37.2MB
hello-world	latest	305243c73457	20 months ago	24.4kB

d) Pousser l'image sur le repo local

```
docker push localhost:5000/firstapp:local
```

```
minikube image load localhost:5000/firstapp:local
```

```
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> docker push localhost:5000/firstapp:local
The push refers to repository [localhost:5000/firstapp]
7c0a2c9ac8d: Pushed
45add851092: Pushed
d674058ff8f: Pushed
a08f362bc9c: Pushed
929017bc7d8: Pushed
dbc1711fe5a: Pushed
95a7cd6e6ae: Pushed
efeb1835bee: Pushed
75e6820e2ab: Pushed
919ce427c94: Pushed
local: digest: sha256:3ff46c40d2e19098eb35b9589b0c91576b0968d705760bdacffca365881ce4a1 size: 856
```

e) Exécuter l'application

```
docker run -d -p 8080:8080 firstapp
```

f) Vérifier que l'application fonctionne

```
winget install curl
```

```
curl http://localhost:8080/
```

7. Déployer l'application dans kube et vérifier

```
kubectl apply -f deployment.yaml
```

```
kubectl get pods
```

```
kubectl get services
```

```
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> kubectl apply -f deployment.yaml
deployment.apps/first-app created
service/first-app-service created
```

```
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
first-app-65dcd57f6b-82g4n	1/1	Running	0	28s
first-app-65dcd57f6b-qn2wt	1/1	Running	0	28s
first-app-65dcd57f6b-tn7kv	1/1	Running	0	28s

```
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> kubectl get services
```

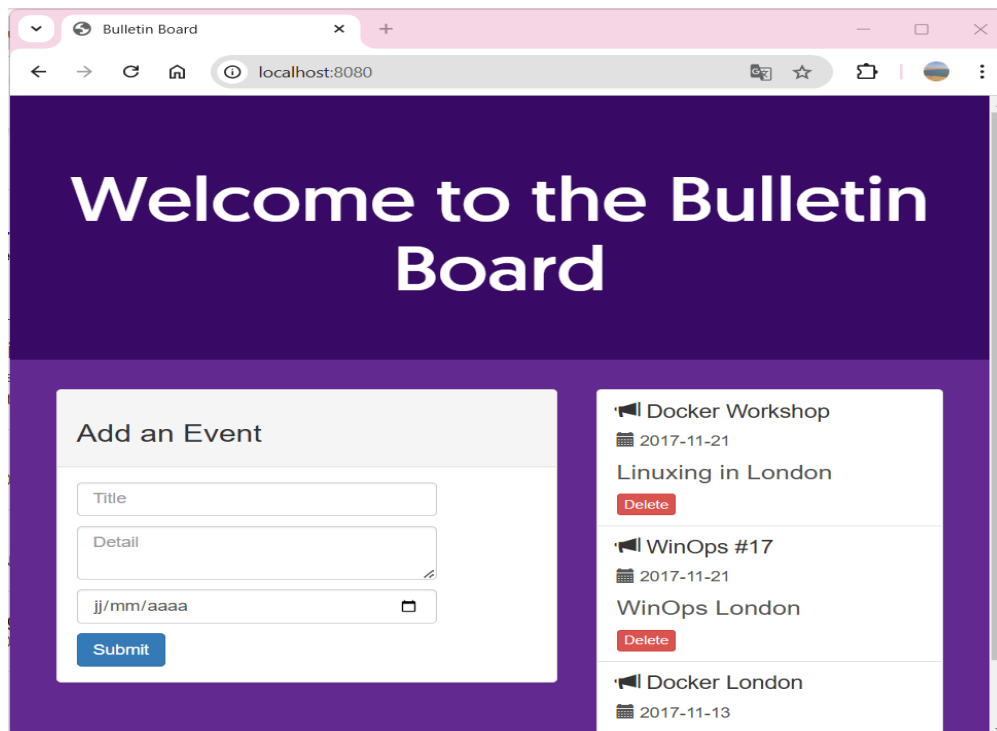
NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
first-app-service	LoadBalancer	10.99.191.123	<pending>	8080:30532/TCP	48s
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	171m

Fichier deployment.yaml pour déployer les conteneurs et les services (conserver l'indentation et mettre dans le même répertoire dockerfile)

```
apiVersion: apps/v1
```

```
kind: Deployment
metadata:
  name: first-app
spec:
  replicas: 3 # Le nombre de réplicas de votre application
  selector:
    matchLabels:
      app: first-app
  template:
    metadata:
      labels:
        app: first-app
    spec:
      containers:
        - name: first-app
          image: localhost:5000/firstapp:local # L'image Docker que nous avons poussée
          ports:
            - containerPort: 3000
---
apiVersion: v1
kind: Service
metadata:
  name: first-app-service
spec:
  selector:
    app: first-app
  ports:
    - protocol: TCP
      port: 8080 # Port externe pour accéder au service
      targetPort: 3000 # Port interne de l'application
  type: LoadBalancer # Expose le service à l'extérieur du cluster
```

Allez dans On peut accéder à l'application par ce lien : <http://localhost:8080/>



g) Afficher les informations sur les services et déploiements.

```
kubectl describe deployments/first-app
kubectl describe services/first-app-service
```

```

PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> kubectl describe deployments/first-app
Name:          first-app
Namespace:     default
CreationTimestamp: Sun, 29 Dec 2024 20:50:41 +0100
Labels:        <none>
Annotations:   deployment.kubernetes.io/revision: 1
Selector:      app=first-app
Replicas:      2 desired | 2 updated | 2 total | 2 available | 0 unavailable
StrategyType:  RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=first-app
  Containers:
    first-app:
      Image:        localhost:5000/firstapp:local
      Port:         3000/TCP
      Host Port:    0/TCP
      Environment:  <none>
      Mounts:       <none>
      Volumes:      <none>
      Node-Selectors:  <none>
      Tolerations:    <none>
Conditions:
  Type           Status  Reason
  ----           -
  Progressing    True    NewReplicaSetAvailable
  Available      True    MinimumReplicasAvailable
OldReplicaSets: <none>
NewReplicaSet:  first-app-65dcd57f6b (2/2 replicas created)
Events:
  Type     Reason              Age   From                  Message
  ----     -
  Normal   ScalingReplicaSet   31m   deployment-controller Scaled up replica set first-app-65dcd57f6b to 5 from 3
  Normal   ScalingReplicaSet   27m   deployment-controller Scaled down replica set first-app-65dcd57f6b to 2 from 5
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app>

```

```
Sélection Administrateur : Windows PowerShell

PS C:\WINDOWS\system32> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
first-app-65dcd57f6b-82g4n          1/1     Running   0           3h3m
first-app-65dcd57f6b-qn2wt          1/1     Running   0           3h3m
first-app-65dcd57f6b-tn7kv          1/1     Running   0           3h3m
PS C:\WINDOWS\system32> kubectl get deployments
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
first-app  3/3     3             3           3h4m
PS C:\WINDOWS\system32> kubectl describe deployments
Name:          first-app
Namespace:     default
CreationTimestamp: Sun, 29 Dec 2024 20:50:41 +0100
Labels:        <none>
Annotations:   deployment.kubernetes.io/revision: 1
Selector:      app=first-app
Replicas:      3 desired | 3 updated | 3 total | 3 available | 0 unavailable
StrategyType:  RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=first-app
  Containers:
    first-app:
      Image:      localhost:5000/firstapp:local
      Port:       3000/TCP
      Host Port:  0/TCP
      Environment: <none>
      Mounts:      <none>
      Volumes:     <none>
      Node-Selectors: <none>
      Tolerations: <none>
Conditions:
  Type           Status  Reason
  ----           -
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets: <none>
NewReplicaSet:  first-app-65dcd57f6b (3/3 replicas created)
Events:         <none>
PS C:\WINDOWS\system32>
```

- h) voir les réplica-sets créé par le déploiement par la commande 'get rs' et changer par la suite le nombre des réplicas sets par la commande 'scale'

```
PS C:\WINDOWS\system32> kubectl get rs
NAME                                DESIRED   CURRENT   READY   AGE
first-app-65dcd57f6b                3         3         3       3h11m
PS C:\WINDOWS\system32>
```

- i) Changer le nombre de réplicas à 5.

```
PS C:\WINDOWS\system32> kubectl get rs
NAME                                DESIRED   CURRENT   READY   AGE
first-app-65dcd57f6b                3         3         3       3h11m
PS C:\WINDOWS\system32> kubectl scale deployments/first-app --replicas=5
deployment.apps/first-app scaled
PS C:\WINDOWS\system32> kubectl get deployments
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
first-app  5/5     5             5           3h17m
PS C:\WINDOWS\system32> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
first-app-65dcd57f6b-82g4n          1/1     Running   0           3h19m
first-app-65dcd57f6b-9fts9          1/1     Running   0           2m26s
first-app-65dcd57f6b-h2z8v          1/1     Running   0           2m26s
first-app-65dcd57f6b-qn2wt          1/1     Running   0           3h19m
first-app-65dcd57f6b-tn7kv          1/1     Running   0           3h19m
PS C:\WINDOWS\system32>
```

- j) Ensuite nous allons réduire le nombre des réplicas à 2.

```
kubectl scale deployments/first-app --replicas=2
```

```

PS C:\WINDOWS\system32> kubectl scale deployments/first-app --replicas=2
deployment.apps/first-app scaled
PS C:\WINDOWS\system32> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
first-app-65dcd57f6b-82g4n         1/1     Running   0           3h21m
first-app-65dcd57f6b-tn7kv         1/1     Running   0           3h21m
PS C:\WINDOWS\system32>

```

k) Afficher les détails sur le services.

```

PS C:\WINDOWS\system32> kubectl describe services/first-app-service
Name:                                first-app-service
Namespace:                           default
Labels:                               <none>
Annotations:                           <none>
Selector:                             app=first-app
Type:                                  LoadBalancer
IP Family Policy:                      SingleStack
IP Families:                           IPv4
IP:                                    10.99.191.123
IPs:                                   10.99.191.123
Port:                                  <unset> 8080/TCP
TargetPort:                            3000/TCP
NodePort:                              <unset> 30532/TCP
Endpoints:                             10.244.0.20:3000,10.244.0.21:3000
Session Affinity:                      None
External Traffic Policy:                Cluster
Events:                                 <none>
PS C:\WINDOWS\system32>

```

l) L'environnement kubernetes dashboard :

minikube dashboard

```

Administrateur : Windows PowerShell
PS C:\WINDOWS\system32> minikube dashboard
* Activation du tableau de bord...
  - Utilisation de l'image docker.io/kubernetesui/metrics-scraper:v1.0.8
  - Utilisation de l'image docker.io/kubernetesui/dashboard:v2.7.0
* Certaines fonctionnalités du tableau de bord nécessitent le module complémentaire metrics-server. Pour activer toutes
les fonctionnalités, veuillez exécuter :

    minikube addons enable metrics-server

* Vérification de l'état du tableau de bord...
* Lancement du proxy...
* Vérification de l'état du proxy...
* Ouverture de http://127.0.0.1:51101/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/
dans votre navigateur par défaut...

```

Kubernetes Dashboard

127.0.0.1:57051/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard/proxy/#/workloads?namespace=default

kubernetes default Recherche

Charges de travail

Workloads

- Cron Jobs
- Daemon Sets
- Deployments
- Jobs
- Pods
- Replica Sets
- Replication Controllers
- Stateful Sets

Service

- Ingresses
- Ingress Classes
- Services

Config and Storage

- Config Maps
- Persistent Volume Claims
- Secrets
- Storage Classes

Cluster

- Cluster Role Bindings
- Cluster Roles
- Events
- Namespaces
- Network Policies

Deployments

Running: 1

Pods

Running: 2

Replica Sets

Running: 1

Nom	Images	Étiquettes	Pods	Date de création
first-app	localhost:5000/firstapp:local	-	2 / 2	3 hours ago

Nom	Images	Étiquettes	Noeud	Statut	Redémarrage	Utilisation CPU (coeurs)	Utilisation mémoire (octets)	Date de création
first-app-65dc57f6b-82g4n	localhost:5000/firstapp:local	app: first-app pod-template-hash: 65dc57f6b	minikube	Running	0	-	-	3 hours ago
first-app-65dc57f6b-tn7kv	localhost:5000/firstapp:local	app: first-app pod-template-hash: 65dc57f6b	minikube	Running	0	-	-	3 hours ago

Nom	Images	Étiquettes	Pods	Date de création
first-app-65dc57f6b	localhost:5000/firstapp:local	app: first-app pod-template-hash: 65dc57f6b	2 / 2	3 hours ago

Cluster > Noeuds

Service

- Ingresses
- Ingress Classes
- Services

Config and Storage

- Config Maps
- Persistent Volume Claims
- Secrets
- Storage Classes

Cluster

- Cluster Role Bindings
- Cluster Roles
- Events
- Namespaces
- Network Policies

Noeuds

Nom	Étiquettes	Prêt	Requêtes CPU (coeurs)	Limites CPU (coeurs)	CPU capacity (coeurs)	Requêtes mémoire (octets)	Limites mémoire (octets)	Memory capacity (bytes)	Pods	Date de création
minikube	beta.kubernetes.io/arch: amd64 beta.kubernetes.io/os: linux kubernetes.io/arch: amd64	True	750,00m (37,50%)	0,00m (0,00%)	2,00	170,00Mi (4,45%)	170,00Mi (4,45%)	3,73Gi	13 (11,82%)	8 hours ago

Charges de travail > Pods

Workloads

- Cron Jobs
- Daemon Sets
- Deployments
- Jobs
- Pods
- Replica Sets

Nom	Images	Étiquettes	Noeud	Statut	Redémarrage	Utilisation CPU (coeurs)	Utilisation mémoire (octets)	Date de création
first-app-65dc57f6b-82g4n	localhost:5000/firstapp:local	app: first-app pod-template-hash: 65dc57f6b	minikube	Running	0	-	-	3 hours ago
first-app-65dc57f6b-tn7kv	localhost:5000/firstapp:local	app: first-app pod-template-hash: 65dc57f6b	minikube	Running	0	-	-	3 hours ago

Service

Workloads

- Cron Jobs
- Daemon Sets
- Deployments
- Jobs
- Pods
- Replica Sets
- Replication Controllers
- Stateful Sets

Service

Nom	Étiquettes	Type	IP cluster	Terminaisons internes	Terminaisons externes	Date de création
first-app-service	-	LoadBalancer	10.99.191.123	first-app-service:8080 TCP first-app-service:30532 TCP	-	3 hours ago
kubernetes	component: apiserver provider: kubernetes	ClusterIP	10.96.0.1	kubernetes:443 TCP kubernetes:0 TCP	-	6 hours ago

Charges de travail > Replica Sets

Workloads

- Cron Jobs
- Daemon Sets
- Deployments
- Jobs
- Pods
- Replica Sets

Nom	Images	Étiquettes	Pods	Date de création
first-app-65dc57f6b	localhost:5000/firstapp:local	app: first-app pod-template-hash: 65dc57f6b	2 / 2	3 hours ago

8. Arrêter les services et déploiements

```
kubectl delete svc first-app-service --namespace=default --force --grace-period=0
kubectl delete deployment firstapp --namespace=default --force --grace-period=0
minikube stop
```

```
Administrateur : Windows PowerShell
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> kubectl delete svc first-app-service --namespace=default --force --grace-period=0
Warning: Immediate deletion does not wait for confirmation that the running resource has been terminated. The resource may continue to run on the cluster indefinitely.
Service "first-app-service" force deleted
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> kubectl delete deployment firstapp --namespace=default --force --grace-period=0
Warning: Immediate deletion does not wait for confirmation that the running resource has been terminated. The resource may continue to run on the cluster indefinitely.
Deployment.apps "first-app" force deleted
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> kubectl get pods
No resources found in default namespace.
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app>
```

9. Démarrez un cluster avec 3 nœuds

1. Création du cluster, en spécifiant les ressources

```
minikube start --nodes 3 -p multinode-node-test --driver=docker --cpus=2 --memory=1500
```

```
Administrateur : Windows PowerShell
PS C:\WINDOWS\system32> minikube start --nodes 3 -p multinode-node-test --driver=docker --cpus=2 --memory=1800
* [multinode-node-test] minikube v1.34.0 sur Microsoft Windows 11 Pro 10.0.26100.2605 Build 26100.2605
* Utilisation du pilote docker basé sur la configuration de l'utilisateur

X Docker Desktop n'a que 7836Mo disponibles, vous pouvez rencontrer des échecs de déploiement d'applications.
* Suggestion :

    1. Ouvrez le menu "Docker Desktop" en cliquant sur l'icône Docker dans la barre d'état système
    2. Cliquez sur "Settings"
    3. Cliquez sur "Ressources"
    4. Augmentez la barre de défilement "Memory" à 2.25 GB ou plus
    5. Cliquez sur "Apply & Restart"
* Documentation: https://docs.docker.com/docker-for-windows/#resources

* Utilisation du pilote Docker Desktop avec le privilège root
* Démarrage du nœud "multinode-node-test" primary control-plane dans le cluster "multinode-node-test"
* Extraction de l'image de base v0.0.45...
* Création de docker container (CPU=2, Memory=1800Mo) ...
! Failing to connect to https://registry.k8s.io/ from inside the minikube container
* Pour extraire de nouvelles images externes, vous devrez peut-être configurer un proxy : https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
* Préparation de Kubernetes v1.31.0 sur Docker 27.2.0...
  - Génération des certificats et des clés
  - Démarrage du plan de contrôle ...
  - Configuration des règles RBAC ...
* Configuration de CNI (Container Networking Interface)...
* Vérification des composants Kubernetes...
  - Utilisation de l'image gcr.io/k8s-minikube/storage-provisioner:v5
* Modules activés: storage-provisioner, default-storageclass

* Démarrage du nœud "multinode-node-test-m02" worker dans le cluster "multinode-node-test"
* Extraction de l'image de base v0.0.45...
* Création de docker container (CPU=2, Memory=1800Mo) ...
* Options de réseau trouvées :
  - NO_PROXY=192.168.49.2
  - NO_PROXY=192.168.49.2
! Failing to connect to https://registry.k8s.io/ from inside the minikube container
* Pour extraire de nouvelles images externes, vous devrez peut-être configurer un proxy : https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
* Préparation de Kubernetes v1.31.0 sur Docker 27.2.0...
  - env NO_PROXY=192.168.49.2
* Vérification des composants Kubernetes...

* Démarrage du nœud "multinode-node-test-m03" worker dans le cluster "multinode-node-test"
* Extraction de l'image de base v0.0.45...
* Création de docker container (CPU=2, Memory=1800Mo) ...
```

Vérification:

```
PS C:\WINDOWS\system32> minikube profile list
|-----|
| Profile | VM Driver | Runtime | IP | Port | Version | Status | Nodes | Active Profile | Active Kubecontext |
|-----|
| multinode-node-test | docker | docker | 192.168.49.2 | 8443 | v1.31.0 | Running | 3 | | * |
|-----|

PS C:\WINDOWS\system32> kubectl get nodes
NAME                 STATUS    ROLES    AGE   VERSION
multinode-node-test  Ready    control-plane   106s   v1.31.0
multinode-node-test-m02  Ready    <none>        66s   v1.31.0
multinode-node-test-m03  Ready    <none>        27s   v1.31.0
```

On peut ajouter des nœuds workers si on veut avec la commande (en fonction de la performance de la machine hôte):

```
minikube node add -p multinode-node-test
```

```
PS C:\Users\PC> kubectl get nodes
NAME                STATUS    ROLES    AGE   VERSION
multinode-node      Ready    control-plane   9h   v1.31.0
multinode-node-m02   Ready    <none>         9h   v1.31.0
multinode-node-m03   Ready    <none>         9h   v1.31.0
PS C:\Users\PC>
```

2. On vérifie l'état de nos nœuds :

minikube status -p multinode-node-test

```
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> minikube status -p multinode-node-test
multinode-node-test
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

multinode-node-test-m02
type: Worker
host: Running
kubelet: Running

multinode-node-test-m03
type: Worker
host: Running
kubelet: Running

PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> kubectl apply -f deployment.yaml
deployment.apps/first-app created
service/first-app-service created
```

11. Traitement d'erreurs

En cas de problème de démarrage des pods comme indiqué dans cette exemple :

```
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> kubectl get pods -o wide
NAME                READY    STATUS              RESTARTS   AGE   IP              NODE                NOMINATED NODE   READINESS GATES
first-app-65dcd57f6b-247ww    0/1      ImagePullBackOff    0          110s  10.244.1.2      multinode-node-test-m02    <none>           <none>
first-app-65dcd57f6b-4qv5r    0/1      ImagePullBackOff    0          110s  10.244.0.3      multinode-node-test    <none>           <none>
first-app-65dcd57f6b-xlr5l    0/1      ErrImagePull        0          110s  10.244.2.2      multinode-node-test-m03    <none>           <none>

PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> kubectl logs first-app-65dcd57f6b-247ww
Error from server (BadRequest): container "first-app" in pod "first-app-65dcd57f6b-247ww" is waiting to start: trying and failing to pull image

PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> kubectl describe pod first-app-65dcd57f6b-247ww
Name:               first-app-65dcd57f6b-247ww
Namespace:          default
Priority:            0
Service Account:    default
Node:               multinode-node-test-m02/192.168.49.3
Start Time:         Tue, 31 Dec 2024 21:45:24 +0100
Labels:             app=first-app
                   pod-template-hash=65dcd57f6b
Annotations:        <none>
Status:             Pending
IP:                10.244.1.2
IPs:               10.244.1.2
Controlled By:      ReplicaSet/first-app-65dcd57f6b
Containers:
  first-app:
    Container ID:
    Image:         localhost:5000/firstapp:local
    Image ID:
    Port:         3000/TCP
    Host Port:    0/TCP
    State:        Waiting
      Reason:     ImagePullBackOff
    Ready:       False
    Restart Count: 0
    Environment: <none>
```

Il faut vérifier que le registry local est fonctionnel, et que l'image de l'application est dans ce registry. N'hésitez pas à refaire les étapes précédentes (création du registre, addons, push, load, deploy et run, ...)

Si vous avez cette erreur :

```
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> docker run -d -p 5000:5000 --name local-registry registry:2
docker: Error response from daemon: Conflict. The container name "/local-registry" is already in use by container "1ce0781fb24b57ddb0175f79d94b5a2853b3c6a645e55c108cf7b47ed7858e78". You have to remove (or rename) that container to be able to reuse that name.
See 'docker run --help'.
```

Supprimer le container associé : avec la commande :

```
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> docker container rm 1ce0781fb24b57ddb0175f79d94b5a2853b3c6a645e55c108cf7b47ed7858e78
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> docker run -d -p 5000:5000 --name local-registry registry:2
fc097b5681afb01177492f534440ac639b7f0d408041957479f0bde25bdd5c0
```

Puis continuer à exécuter la suite des commandes :

```
docker run -d -p 5000:5000 --name local-registry registry:2
minikube addons enable registry -p multinode-node-test
docker push localhost:5000/firstapp:local
minikube image load localhost:5000/firstapp:local -p multinode-node-test
kubectl apply -f deployment.yaml
kubectl get pods -o wide
```

```
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> kubectl get pods -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP              NODE                                NOMINATED NODE   READINESS GATES
first-app-65dcd57f6b-247ww          1/1     Running   0           89m   10.244.1.2      multinode-node-test-m02            <none>            <none>
first-app-65dcd57f6b-4qv5r          1/1     Running   0           89m   10.244.0.3      multinode-node-test                <none>            <none>
first-app-65dcd57f6b-xlr5l          1/1     Running   0           89m   10.244.2.2      multinode-node-test-m03            <none>            <none>
```

Ici chaque replica s'exécute sur un node.

<http://localhost:8080/>

12. Supprimer un node

```
kubectl delete nodes multinode-node-test-m03 --context=multinode-node-test
```

```
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> kubectl delete nodes multinode-node-test-m03 --context=multinode-node-test
node "multinode-node-test-m03" deleted
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> kubectl --context=multinode-node-test get nodes
NAME                                STATUS    ROLES    AGE   VERSION
multinode-node-test                Ready     control-plane   3h6m   v1.31.0
multinode-node-test-m02            Ready     <none>         3h5m   v1.31.0
```

m) Supprimer un cluster

```
minikube delete -p multinode-node-test
```

```
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> minikube profile list
-----
| Profile | VM Driver | Runtime | IP       | Port | Version | Status | Nodes | Active Profile | Active Kubecontext |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| multinode-node | docker | docker | 192.168.49.2 | 8443 | v1.31.0 | Running | 3 | | *
```

```
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> minikube delete -p multinode-node
* Suppression de "multinode-node" dans docker...
* Suppression du conteneur "multinode-node" ...
* Suppression du conteneur "multinode-node-m02" ...
* Suppression du conteneur "multinode-node-m03" ...
* Suppression du répertoire C:\Users\PC\.minikube\machines\multinode-node...
* Suppression du répertoire C:\Users\PC\.minikube\machines\multinode-node-m02...
* Suppression du répertoire C:\Users\PC\.minikube\machines\multinode-node-m03...
* Le cluster "multinode-node" a été supprimé.
PS C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app>
```